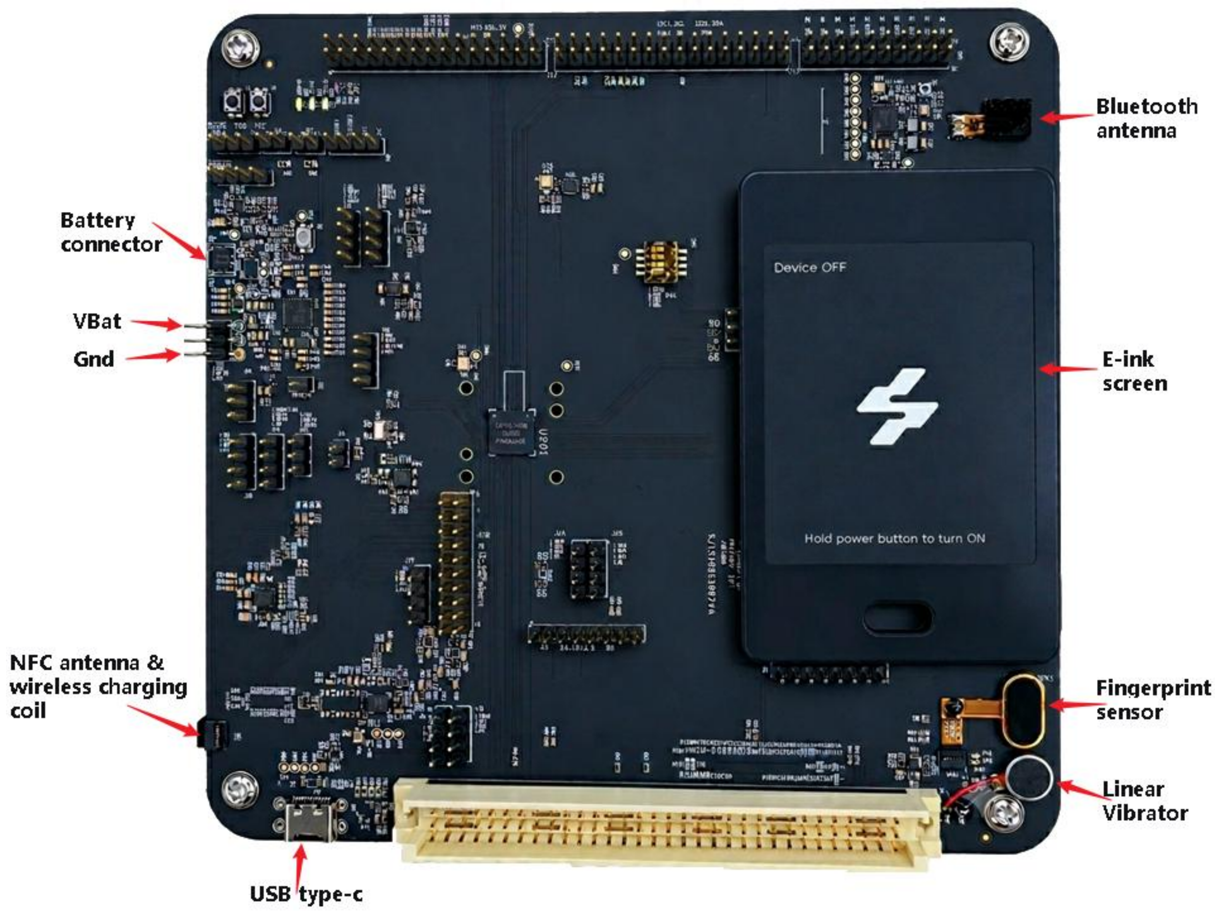
# NTO EVB Quick Start Guide

## 1. Overview

The NTO EVB is an evaluation board based on the Daric NTO SoC. It integrates a rich set of peripherals for development and validation purposes:



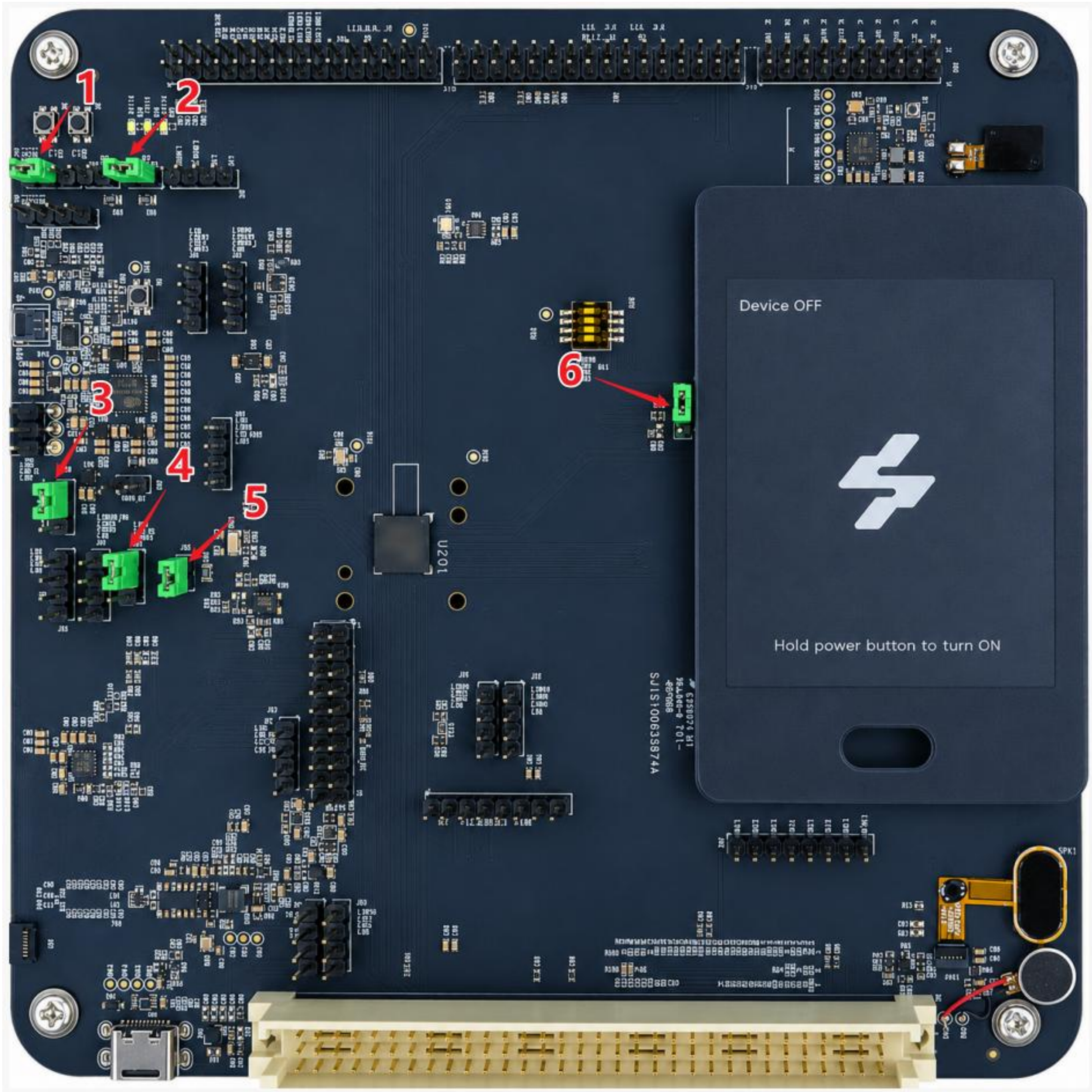

| Peripheral | Interface | Notes |
|---|---|---|
| E-Ink LCM | SPI (SPIM1) | Ed028tc1 module<br>VBAT-powered LDO |
| NFC | SPI (SPIM2) + I2C0 | CIU98N60A2 moduel<br>VBAT-powered |
| Wireless Charging | I2C1 (PA0/PA1) | NU1680QDHB module |
| Bluetooth | UART2 (PB13/PB14) | BT8932B module<br>VSYS-powered LDO |
| Fingerprint Sensor | SPI (SPIM3) | VSYS-powered LDO |

| | | |
|---|---|---|
| Vibrator (Haptic feedback) | I2C2 (PE10/PE11) | AW86234 module<br>VBAT-powered |
| RGB LED Driver | I2C2 (PE10/PE11) | AW2033 module<br>VBAT-powered |
| QSPI Flash | SPI (SPIM0) | Winbond 1 Gbit |
| Touch Panel | I2C3 (PC5/PC6) | CST9220 module |
| USB Type-C | USB 2.0 HS | VBUS AW32102 load switch |
| JTAG / SWD | J302 / JTAG1 | Debug & programming |
| DUART | PA3 (RX) / PA4 (TX) | Serial debug console |

---

## 2. Jumper Cap Configuration

The EVB has six jumper cap positions that configure key signal routing and power measurement points. Their locations are marked in the photo below (numbers 1–6 correspond to the red labels on the board image).




| No. | Connector | Function | Default Setting |
|---|---|---|---|
| 1 | J108 | Reset source selection for the Daric SoC. Selects which signal drives the system reset. | Daric `PG_NRST_SYS` pin selected |
| 2 | J617 | Current measurement point for the TG28 PMIC-output VDD3V3 | Jumper shorted (normal operation) |

| | | | |
|---|---|---|---|
| | | rail. Inserting an ammeter here (with R121 adjusted to an appropriate sense resistor value) allows power consumption measurement on this rail. | |
| 3 | J109 | TG28 PIN32 (GPIO/FB5/RTCLDO2) signal selection. Connects this pin to either VDCDC5 or GND. | `VDCDC5` selected |
| 4 | J111 | TG28 PIN13 (BLDOIN) signal selection. Connects this pin to either VSYS or VDD3V3. | `VSYS` selected |
| 5 | J925 | Current measurement point for the Daric SoC VDDAO power rail. Inserting an ammeter here (with R978 adjusted to an appropriate sense resistor value) allows power consumption measurement on this rail. | Jumper shorted (normal operation) |
| 6 | J201 | Daric VDDRR voltage selection. Connects the VDDRR supply to either TG28-output VDDRR0_1 or VDD3V3. | `VDDRR0_1` (TG28 output) selected |

Jumpers J617 (VDD3V3) and J925 (VDDAO) are designed as in-line current measurement points:

1. Remove the jumper cap.
2. Adjust the series resistor (`R121` for J617, `R978` for J925) to a suitable low-value sense resistor.
3. Connect an ammeter or current probe across the two header pins.

4. Measure the rail current during operation.

## 3. Power Supply Methods

The EVB supports three power supply methods. Understanding the characteristics and limitations of each method is essential before use.

### 3.1 Method 1 — Battery via CON101

Connect a single-cell LiPo/Li-ion battery to the DF58-2P-1.2V(21) connector (`CON101`):

- Pin 1: Battery_CELL+ (VBAT)
- Pin 2: Battery_CELL- (GND)

Battery Protection IC Activation (Required on First Use and After Each Battery Removal)

The battery includes a protection IC that must be activated before the board can power on. The activation procedure is as follows:

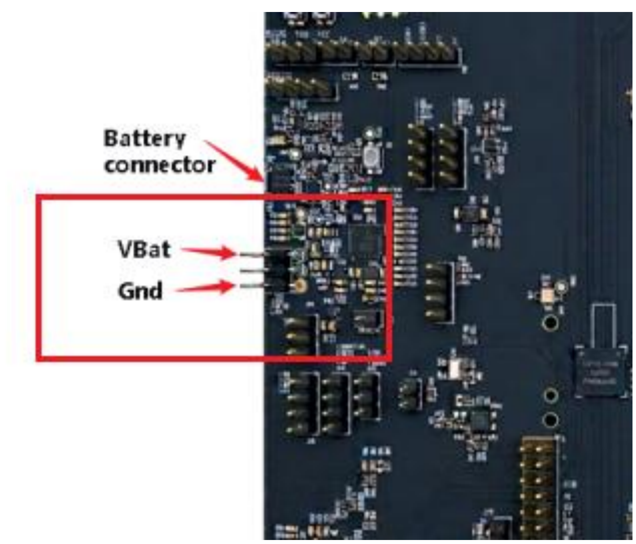
1. Connect the battery to CON101.
2. Insert a USB Type-C cable into J401 (connect to a PC or USB charger).
3. Wait for the E-Ink LCM display to show a change — this confirms the protection IC has been activated.
4. The board can now be powered on normally using the Power Key.

This activation must be repeated every time the battery is physically disconnected and reconnected. Without activation, pressing the Power Key will have no effect.

---

### 3.2 Method 2 — External Adjustable Power Supply via Header Pins

Connect a bench power supply to the external power header :




- Pin 1: `VBAT` (positive)
- Pin 2: `GND` (negative)

Required voltage: Set the output to exactly 4.2 V before connecting to the board.

Steps:

1. Set the adjustable power supply output to 4.2V with current limiting set appropriately (e.g., 500 mA).
2. Verify the polarity — connect positive to `VBAT` and negative to `GND`.
3. Enable the power supply output.
4. The board can now be powered on using the Power Key.

This method provides the same full-peripheral capability as a battery. It is convenient for current measurement and power analysis during development. No activation step is needed.

---

### 3.3 Method 3 — USB Type-C ( Limited Functionality)

Connect a standard USB Type-C cable from a PC or USB power adapter to connector.

USB power has two use cases:

1. Battery Protection IC Activation — When a battery is connected but not yet activated, inserting USB will trigger the activation sequence (see Method 1 above).
2. System Boot (SoC Only) — USB can supply enough power to boot the Daric SoC and run on-chip functionality.

If the system is booted on USB power alone (no battery connected), the following peripherals will NOT function correctly, as they require a dedicated VBAT supply:

- Eink LCM

- Vibrator / Haptic driver

- RGB led module

- NFC

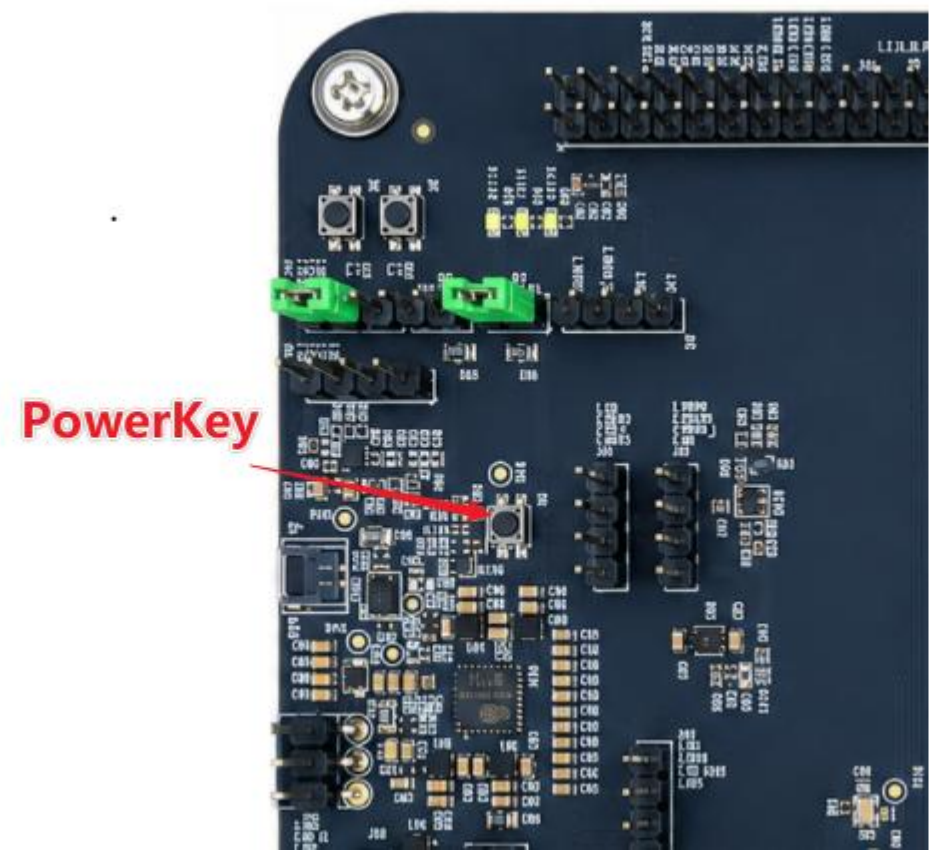
USB power is suitable for debugging Daric on-chip features and firmware download, but not for full peripheral validation.

---

## 4. Powering On and Off

### 4.1 Power On

1. Ensure the power supply is connected and active (see Section 3).
2. Press and hold the Power Key (SW3) for approximately 1.5 seconds, then release.




3. If the system boots successfully, the E-Ink LCM will display the boot screen within a few seconds.

If the E-Ink screen does not update or shows an abnormal image, connect a DUART console (see Section 6) to observe the boot log.

**4.2 Normal Power Off**

Press and hold `SW3` until the system shuts down gracefully (behavior depends on firmware implementation).

**4.3 Force Shutdown (System Hung)**

If the system becomes unresponsive and cannot be shut down normally:

Press and hold SW3 for at least 10 seconds  to force a hardware power-off.

---

## 5. Board Layout & Key Components

| **Reference** | **Component / Function** |
|---|---|
| CON101 | Battery connector (DF58-2P-1.2V(21)) |
| J401 | USB Type-C connector |

| | |
|---|---|
| SW3 | Power Key |
| SW2 / SW3 | Reset |
| CON701 | E-Ink LCM BTB 20-pin connector |
| J901 | Fingerprint sensor ZIF connector |
| CON801 | Touch panel ZIF connector |
| U101 | PMIC (TG28) |
| U105 | Charge IC (ETA4662) |
| U201 | Daric NTO SoC (BGA188) |
| U920 | Wireless charging IC (NU1680) |
| U715 | NFC IC (CIU98N60A2) |
| U921 | Bluetooth module (BT8932B) |
| J301 / J302 | DUART / SWD debug headers |
| JTAG1 | JTAG 20-pin debug connector |
| J607 | 30-pin GPIO expansion header |
| J601A/B/C | 96-pin socket (board-to-board expansion) |

PE5 is exposed on the GPIO expansion header (J607). This pin is used for forcing firmware download mode (see Section 8).

---

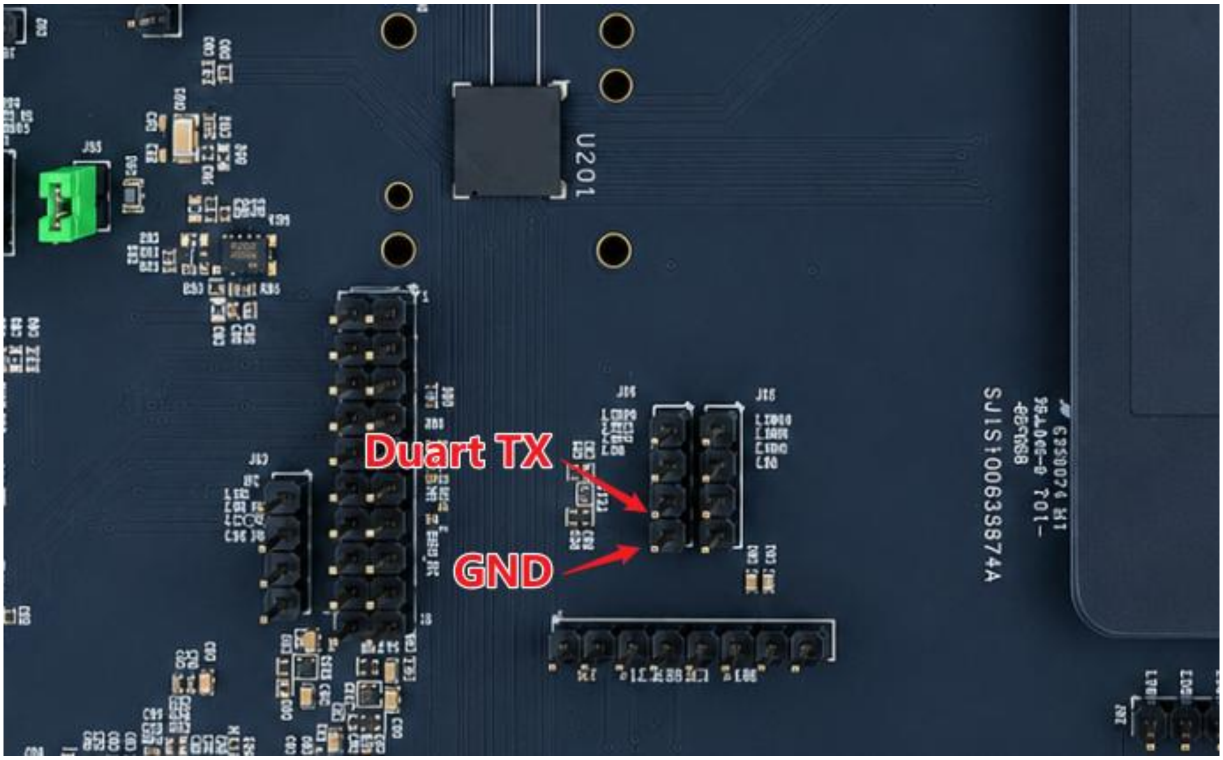
## 6. Debug via DUART Console

The EVB exposes a DUART (serial debug console) for monitoring boot logs from BootROM, Bootloader, and Firmware stages.

### 6.1 Hardware Connection

Connect a USB-to-UART adapter (3.3 V logic level) to the DUART header:

| EVB Signal | Pin | Direction |
|---|---|---|
| DUART TX (PA4) | J301 Pin 3 | EVB → Host |
| GND | J301 Pin 4 | Common ground |




Refer to J301 (4-pin header, labeled VDDPAD / XRSTN / DUART / GND) in the schematic.

### 6.2 Terminal Settings

| Parameter | Value |
|---|---|
| Baud Rate | **921600** |
| Data Bits | 8 |
| Parity | None |
| Stop Bits | 1 |
| Flow Control | None |

### 6.3 Expected Boot Log Sequence

[BootROM] → SoC internal ROM initialization

[Bootloader] → Hardware init, PMIC setup, power-on type checks

[Firmware] → OS / application initialization

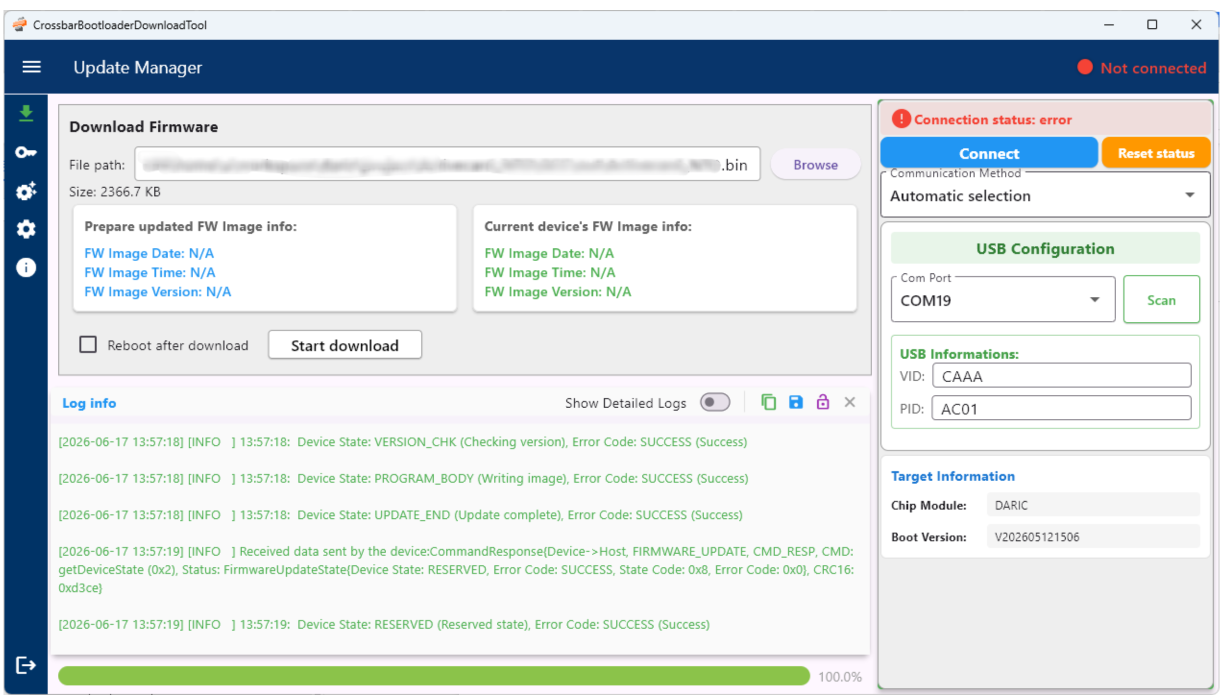
```
[        0 ms][      SYS] Hello BootROM V202605111434 for DARIC built at May 11 2026 14:35:13 !
[        2 ms][      SYS] Config MPU...
[        3 ms][      SYS] MPU Enabled.
[        4 ms][      SYS] ICache Enabled.
[        4 ms][      SYS] Init PMIC I2C1 Interface...
[        5 ms][      SYS] Configuring PMIC TG28...
[       21 ms][      SYS] Done.
[       21 ms][      SYS] Set clock to 700MHz
[        1 ms][      SYS] Reset SysTick due to change of clock
[        0 ms][      SYS] Check the app image...
[       32 ms][      SYS] The app image is good, jump to the app!
[0 ms][BOOT] start bootloader, information: V202605121506, Build @ May 12 2026 15:57:00.
[0 ms][BOOT] update bootloader version
[0 ms][BOOT] init peripherals
[0 ms][BOOT] HAL init
[0 ms][BOOT] HAL PINMAP init
[0 ms][BOOT] set clock to 400MHZ
[0 ms][BOOT] init gpio NoPull
[1 ms][BOOT] Download GPIO port init
[1 ms][BOOT] tg28 init
[8 ms][BOOT] get fastboot flag
[8 ms][BOOT] get fastboot flag from AO, sleepmode: 4, bootmode: 4
[9 ms][BOOT] get_fastboot_flag, is fastboot: 0
[9 ms][BOOT] init ReRAM file system
[10 ms][BOOT] init NAND flash file system
[76 ms][BOOT] get secure state
[77 ms][BOOT] read_fdl_pin Enter Download mode: 0
[114 ms][FOTA] fota avaliable: 0
[114 ms][BOOT] bootloader boot mode
[115 ms][BOOT] secure feature disabled, jump to app!
[115 ms][BOOT] ReRAM firmware header: 0x61002000, 0x6009dd15
[116 ms][BOOT] Jump to firmware application
```

If the system hangs at the BootROM or Bootloader stage, check the power supply voltage and verify that the correct firmware has been flashed.

---

## 7. Firmware Download

Use the **CrossbarBootloaderDownloadTool** (provided separately) to download firmware to the EVB over USB.



### 7.1 Prerequisites

- USB Type-C cable connected between the EVB (`J401`) and the host PC.
- CrossbarBootloaderDownloadTool installed on the host PC.
- Firmware binary (`.bin`) ready.

### 7.2 Download Procedure

1. Open **CrossbarBootloaderDownloadTool** on the host PC.
2. Load the firmware file in the tool.
3. Click the "Connect" button in the tool — do this before powering on or resetting the board.
4. Trigger the EVB to enter Firmware Download Mode (see Section 8).
5. Once the tool shows "Connected" status, release the Power Key (if using Method A below).
6. The tool will automatically start the firmware download process.
7. Wait for the download to complete. Do not disconnect USB during this process.
8. After a successful download, the EVB will reboot automatically or press `SW3` to restart.

Always click the "Connect" button in the tool before triggering download mode on the EVB.

---

## 8. Entering Firmware Download Mode

The Bootloader checks two conditions at startup to decide whether to enter Firmware Download Mode. Either condition is sufficient:

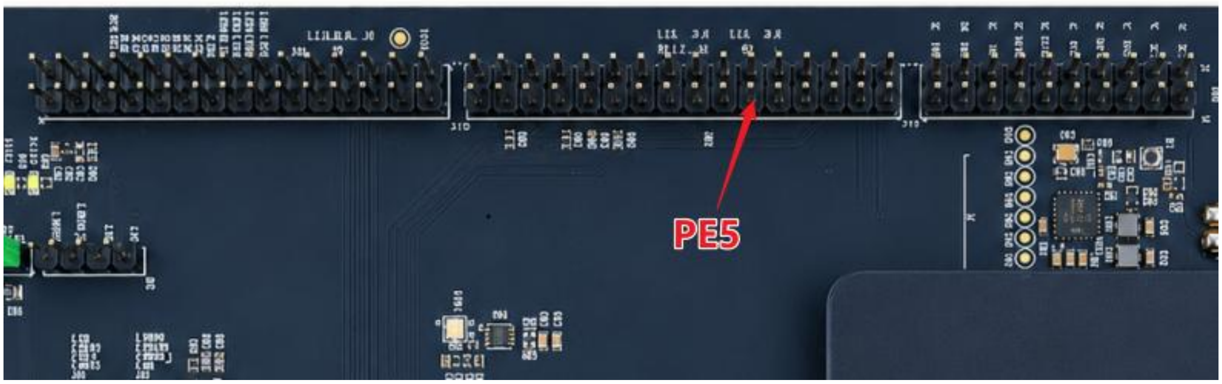
| Condition | Method | Notes |
|---|---|---|
| Power Key held down | Keep SW3 pressed during power-on or reboot | Release only after the tool shows "Connected" |
| GPIO PE5 pulled to GND | Short PE5 to GND with a jumper wire | Every subsequent boot will auto-enter download mode while shorted |

### 8.1 Method A — Power Key Method (One-Time)

1. Click "Connect" in CrossbarBootloaderDownloadTool.
2. Power on the EVB (or press reset) while holding `SW1` down.
3. Plugin usb cable.
4. When the tool shows "Connected", release `SW3`.
5. Firmware download begins automatically.

### 8.2 Method B — PE5 to GND Method (Persistent)

1. Use a dupont jumper wire to short `PE5` to `GND` on the GPIO expansion header (`J607`).




2. Click "Connect" in CrossbarBootloaderDownloadTool.
3. Plugin usb cable.
4. It will automatically enter download mode.
5. After firmware download is complete, remove the jumper to restore normal boot behavior.

---

## 9. Troubleshooting

| Symptom | Possible Cause | Action |
|---|---|---|
| No display on E-Ink screen after power-on | Boot failed / firmware issue | Connect DUART, observe boot log |
| Board does not power on after holding `SW3` | Battery protection IC not activated | Insert USB to activate the protection IC first (see Section 3, Method 1) |
| Board does not power on after holding `SW3` | Insufficient supply voltage | Check battery voltage (≥ 3.5 V) or verify power supply is set to 4.2 V |
| CrossbarBootloaderDownloadTool cannot connect | USB not detected / download mode not triggered | Ensure USB is connected, click "Connect" first, then hold `SW3` during boot |
| Firmware download starts but fails | USB cable quality | Use a high-quality USB cable |
| Peripherals (NFC, BLE, FP, Vibrator) not working | USB-only power supply | Switch to battery or external adjustable supply (Method 1 or 2) |
| Board continuously enters download mode | PE5 is shorted to GND | Remove the jumper wire on `PE5` |
| E-Ink shows garbled image | Display driver issue or power supply glitch | Check LCD voltage & TSC parameter |
| System is unresponsive / hung | Firmware crash | Press and hold `SW3` for at least 10 seconds to force |

| | | |
|---|---|---|
| | | shutdown |